

APPENDIX 3.1: ARCHITECTURAL REGULATIONS OF THE PERSONAL AI COMPANION (TRANSCEPTOR PERSONAL AI COMPANION)

Specification of the personal decentralized Edge AI module, TEE environment, biometric compliance mechanisms, and cognitive mediation protocols under Whitepaper v3.2 and v3.4-Adaptive standards.

1. Concept and Institutional Status of the Personal AI Companion

The Personal AI Companion (Personal AI Companion / Edge AI) is an autonomous, decentralized, and hardware-isolated artificial intelligence module functioning directly on the user's client device within a Trusted Execution Environment (TEE / Secure Enclave / Strict PRM). Within the global Transceptor architecture, the AI Companion is endowed with the legal and algorithmic status of a personal digital advocate, analyst, and sovereign representative of the citizen.
The primary objective of the Edge AI architecture is the absolute exclusion of the analog human factor, central censorship, administrative pressure, corruption, and third parties from moderation, governance, and the execution of legally binding civil actions.

Functional Core Duties

- **Digital Advocate:** Performs preventive auditing of all incoming and outgoing smart contracts, validates transaction compliance against Whitepaper invariants, and protects residents from electoral or fiscal fraud.
- **Personal Analyst:** Executes semantic decompilation of legislative and municipal initiatives, aggregates reputation indices, and models the mathematical consequences of citizen voting.
- **Sovereign Representative:** Generates Zero-Knowledge proofs (ZK-SNARKs) to verify competence and voting rights without transmitting raw biometrics or personal identifiers to external networks.

2. Lightweight Core and Lazy Loading Architecture

To prevent battery degradation, optimize processing overhead, and eliminate thermal throttling on client devices, the system core operates on a strict separation of functional layers.

Background Lightweight Core

The background process of the Lightweight Core is fully isolated from heavy external libraries and third-party software dependencies. In standby mode, the core continuously executes a cyclic Ping-Heartbeat (transmitting encrypted stability attestations to the TEE every 100 milliseconds with a drift limit <= 1ms) and monitors hardware-spatial telemetry.

Dynamic Lazy Loading

Peripheral functional modules utilize dynamic, on-demand loading mechanisms:

1. **1. Initialization:** Heavy peripheral modules (Fin-Companion, Municipal-Companion, Academic-Companion) reside in encrypted storage and do not occupy system RAM.
2. **2. Activation:** At the launch of a user session, the required module is mounted directly into the hardware TEE enclave strictly on demand.
3. **3. Atomic Purge (Zeroization):** Upon session completion or timeout (180 seconds of inactivity), peripheral modules are purged from RAM, and memory sectors are overwritten with zeros (0x00) to prevent memory dump attacks.

3. OPSEC Timelocks and Biometric-Tethering (Fail-Safe Defense)

User interaction with the AI Companion relies on continuous biometric tethering (Biometric-Tethering), ensuring that legally binding decisions are performed exclusively by the authenticated DID passport owner.

OPSEC Invariants ("Zaslon" Defense Circuit)

- **Contactless Biometric Verification:** The Edge AI agent analyzes cornea glare photogrammetry and micro-pupil movements within the smartphone's Secure Enclave.
- **Standby Lock (5 Seconds):** Losing biometric eye contact for 5 seconds triggers an immediate UI mask (Standby Lock), causing host screen captures to return empty byte arrays.
- **Inactivity Timeout (180 Seconds):** Complete absence of biometric presence for 180 seconds triggers an automatic memory purge, wiping allocated session keys.
- **Key Destruction (Zeroization < 10ms):** In the event of physical breach detection or a SIGSEGV system command, CPU cache master decryption keys are erased within 10 milliseconds.

4. Zero-Knowledge Proofs (ZK-SNARKs) and Privacy Architecture

To eliminate identity leaks and electoral profiling, the AI Companion executes privacy protocols based on Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (ZK-SNARKs) under the principle of "Internal Blindness".

- **Verification without Exposure:** The companion generates mathematical proofs confirming that the user possesses valid civil rights, location proofs, or competence scores without disclosing legal names, exact physical coordinates, or raw academic credentials.
- **Identity Isolation:** User identity remains concealed behind anonymized hashes (did_hash), ensuring complete protection against central administrative or political coercion.

5. Cognitive Mediation Protocol (v3.4-Adaptive AI-Architect)

Under the v3.4-Adaptive modernization framework, the AI Companion evolves from a passive filtering agent into an active "Constructive Transformer" (AI Architect).

Transition from Censor to AI Architect

- **Decompilation & Construction:** When a citizen submits an unformalized or non-standard civil initiative without an established feasibility study (Psi = 0), the AI Companion does not reject it outright.
- **Layer Generation:** The semantic engine automatically builds missing required layers: Legal (Comp_law), Financial (Comp_fin), and Ecological (Comp_eco).
- **ScenarioReport Assembly:** The companion delivers a complete, structured ScenarioBrief back to the author for one-click approval via the Personal Companion prior to re-submitting to the Agora.

6. Gradient Adaptation Framework ("Physics vs Failure")

To protect users from unearned penalties during environmental or hardware disruptions, the AI Companion incorporates a two-level gradient response:

- **Pending Buffer Activation:** When sensor telemetry detects anomalies (|M_fact - M_target| > Threshold), the system avoids immediate reputation degradation (delta_W_user = -4.99) and places the state into PENDING.
- **Environmental Correlation (K_env):** The Edge AI agent conducts cross-triangulation with adjacent IoT nodes and sensors to calculate the Environmental Correlation Coefficient (K_env).

Failure Mitigation: If physical interference, severe weather, or signal jamming is verified (K_env >= Threshold_env), an adaptation buffer is granted without applying negative reputation adjustments to the user.